

BrainiBeats: A dual brain-computer interface for musical composition using inter-brain synchrony and emotional valence

Caterina Ceccato
Tilburg University
The Netherlands
c.ceccato@tilburguniversity.edu

Ethel Pruss
Tilburg University
The Netherlands
e.pruss@tilburguniversity.edu

Anita Vrins
Tilburg University
The Netherlands
a.m.vrins@tilburguniversity.edu

Jos Prinsen
Tilburg University
The Netherlands
j.m.prinsen@tilburguniversity.edu

Maryam Alimardani
Tilburg University
The Netherlands
m.alimardani@tilburguniversity.edu

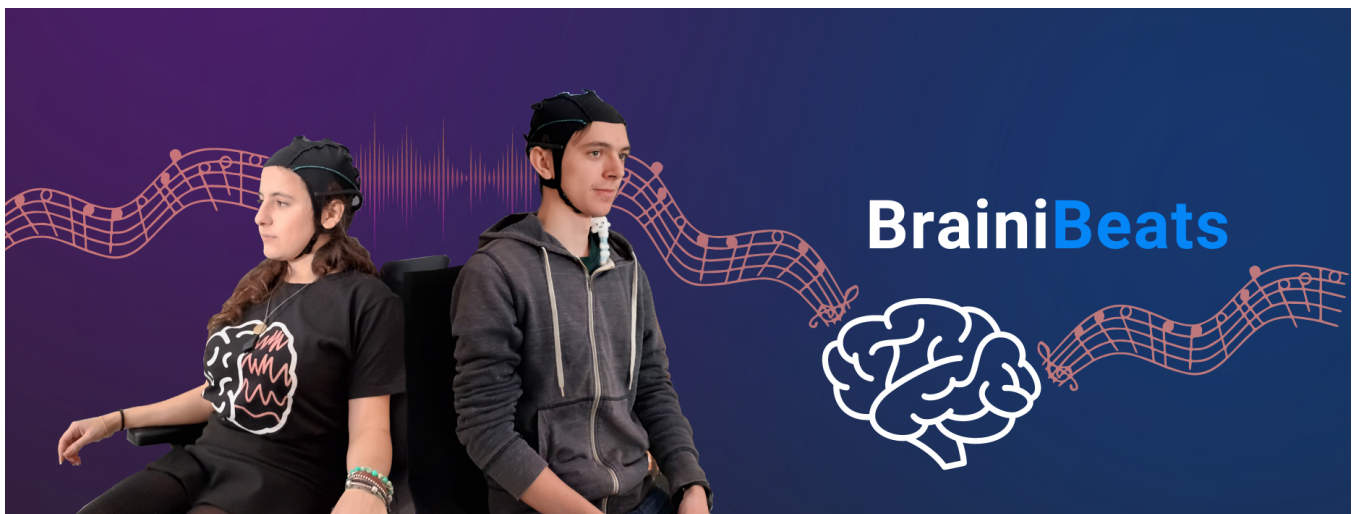


Figure 1: A visualization of two users using the dual-brain BCI system for music generation. BrainiBeats produces note-by-note music based on the amplitude, emotional valence, and synchrony in the EEG brain activity of two users.

ABSTRACT

A dual brain-computer interface (BCI) was developed to translate emotions and synchrony between two users into music. Using EEG signals of two individuals, the system generates live music note-by-note and controls musical parameters, such as pitch, intensity and interval. The users' mean EEG amplitude determines the notes, and their emotional valence modulates the intensity (i.e. volume of music). Additionally, inter-brain synchrony is used to manipulate the interval between notes, with higher synchrony producing more pleasant music and lower synchrony producing less pleasant music. Further research is needed to test the system in an experimental setting, however, literature suggests that neurofeedback based on inter-brain synchrony and emotional valence could be used to

promote positive aspects of group dynamics and mutual emotional understanding.

CCS CONCEPTS

• **Human-centered computing** → **Sound-based input / output; User models; User interface design.**

KEYWORDS

brain-computer interface (BCI), neurofeedback, music, inter-brain synchrony, emotion

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585910>

ACM Reference Format:

Caterina Ceccato, Ethel Pruss, Anita Vrins, Jos Prinsen, and Maryam Alimardani. 2023. BrainiBeats: A dual brain-computer interface for musical composition using inter-brain synchrony and emotional valence. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585910>

1 INTRODUCTION

Music has a profound ability to connect and synchronize people [24, 26]. Participating in music-related tasks can subconsciously increase coordination between individuals, even when there is no explicit intention to do so [13]. Recent studies involving multiple participants, known as hyperscanning, have revealed neural patterns underlying coordination and cooperation during musical activities [2, 26]. For instance, inter-brain synchronization (IBS) has been observed in various cooperative musical tasks, such as singing [40], playing instruments [32, 40] and rhythmic movements [28]. This demonstrates the impact of music on the social brain, as an increase in interpersonal synchronization can lead to increased bonding, group cohesion, and prosocial behavior [52].

While music can impact human physiology, the reverse is also possible by translating physiological signals into musical rhythms. The creation of musical pieces composed from physiological measures such as heart rate, motion-tracking, and galvanic skin response has been explored for educational, therapeutic, and artistic purposes [39, 53]. Traditionally, these musical installations were applied to single users and utilized non-neural measures [53]. However, with the increasing accessibility of EEG equipment and real-time signal-processing algorithms, Brain-Computer Musical Interfaces (BCMI) have created the possibility for neural-activity-based music composition [37].

Two main paradigms for BCMI can be identified from the literature: active and passive composition systems [53]. While active BCMI allows for more experienced musicians to take control of their compositions using, for example, P300 or SSVEP paradigms [43, 50], passive BCMI makes use of mapping systems to transform brain signals into musical output without any knowledge of musical theory or conscious involvement [53]. For instance, a passive BCMI can classify a user's emotions into melodies that match the user's mood or help them with emotion regulation [12, 17]. Such BCMI can be utilized in multi-user settings as well, where the music adapts to the emotional states of two users, e.g., a performer and a listener [15].

The implementation of multi-brain brain-computer interfaces is already observed in various domains including art and entertainment [41, 49]. However, with regards to music production, the attempts are limited to BCMI that are individually controlled by two or more musicians [31]. Despite the evidence from hyperscanning studies that suggest shared emotional experience [6, 38] and inter-brain synchronization [42, 45] during collaborative musical tasks, to date, no study has employed these metrics for development of a multi-user passive BCMI.

The project outlined in this paper is a proof of concept of a passive BCMI system, which uses scale-free rules of sonification, lateralization of brain activity, and inter-brain synchrony between two users to generate note-by-note uniquely co-composed musical melodies from real-time brain signals. Brain activity was collected from two users simultaneously using two wireless EEG headsets and analyzed in real-time to extract features that translated the mental states of the users to music, with a focus on dual brain synchrony and the individual's emotional valence. The proposed system measures three EEG signal characteristics (mean amplitude, frontal alpha asymmetry (FAA), and inter-brain phase lock value

(PLV)) and maps them to the pitch, intensity, and consonance of the produced music.

2 BACKGROUND

The primary aim of mapping EEG activity to music is to represent the user's mental state in a novel way, which requires finding relationships between music and brain activity. The most common approach to do this entails classifying a person's emotional arousal or valence from their EEG signals, and then matching it to pre-generated music that has been rated with similar arousal or valence by listeners [4, 22, 56]. A less explored approach is one that actively generates music based on the individual's brain activity by translating features of EEG signals to musical features that are thought to represent similar concepts. For example, changes in musical intensity have been linked to changes in participants' emotional response [4]. Incremental conversion of neural activity to music is usually achieved by analysis of a single participant's EEG data based on spectral features [23] or scale-free conversions derived from the power-law scaling found in both fluctuations of EEG amplitude and various musical parameters such as pitch and melodic intervals [34, 54]. This paper aims to bring together these previous approaches and combine EEG spectral, temporal, and scale-free features to generate music in a novel live ensemble scenario. The significance of this work lies in the production of ensemble music using BCMI, where the brain activity of two users is translated into unique and expressive music in real-time, note by note.

2.1 Musical Composition and Brain Activity

Music composition is the process of forming a piece of music by combining different elements of music. There are several key elements that impact how the music sounds. The starting point of generating live music is finding the individual notes to be played. For this, the pitch of the music needs to be defined. Pitch relates to the arrangement of notes based on frequency. Perceptually, it modulates whether the sound is experienced as "lower" or "higher". Interestingly, music and neural activity show similar patterns of distribution in terms of frequency. Both music -that is rated as pleasant- and EEG activity follow a Zipfian power-law distribution [18, 34], meaning that there is an inverse relationship between frequency and the amount of observed activity. This relationship can be used to derive a scale-free conversion rule, which has been pioneered in the field of algorithmic music [7, 19] as well as being demonstrated in a few BCMI applications [54]. Using the conversion rule, the mean amplitude of EEG signals can be used to find a corresponding pitch, which determines the note that will be played.

Combining individual notes randomly is not sufficient to generate pleasant-sounding music. Musical theory describes which notes sound pleasant when played simultaneously or in succession. This is determined by the difference in pitch between the two notes, known as the interval. Depending on the interval, the music can either sound consonant or dissonant, which in turn results in a range of perceptual qualities that are often simplified as pleasantness. By analyzing the brain activity of two people, an interesting new feature emerges that can be mapped to musical harmony: inter-brain synchrony. While synchrony within different regions of one participant's brain has been used in BCMI studies

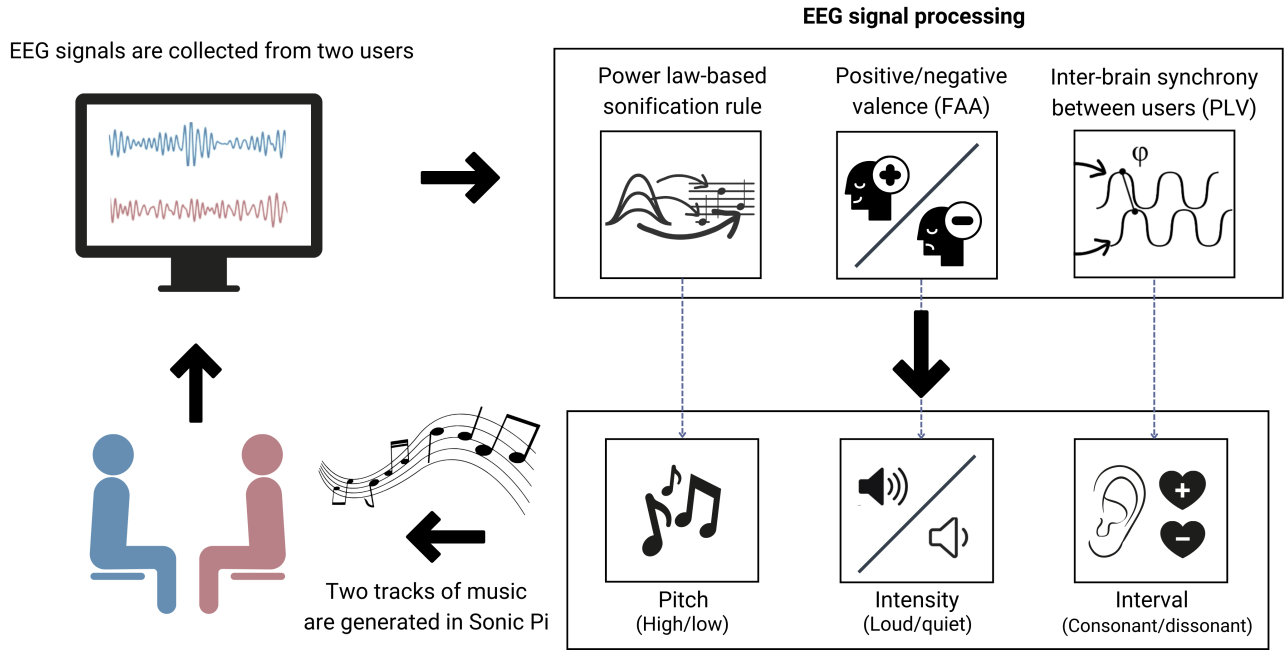


Figure 2: A visual overview of the dual brain BCMI system for live ensemble music production. Three features were extracted from two users’ EEG signals, representing the scale-free structure, emotional valence, and inter-brain synchrony between the users. These features were then mapped to the pitch, intensity, and interval of the generated music. The pitch determines the frequency of notes, intensity modulates the volume of the music, and interval (i.e., the distance between the notes) determines how pleasant the music sounds. These musical features were sent to Sonic Pi where two tracks of music were compiled using Python code (one for each participant), which were then played simultaneously as an ensemble.

before [54], the synchrony between two people has never been explored in a musical context. The most common way of measuring inter-brain synchrony between two people is the phase locking value (PLV), which measures how two signals coming from two brains are phase-locked in a certain time window [11]. The PLV varies between 0 (random phase) to 1 (perfectly synchronized) and is used as a method for connectivity analysis in hyperscanning studies [30, 48]. Previously, PLV has been found to correlate with interpersonal synchrony in various social and collaborative settings [14, 20, 55]. However, there is a lack of studies reporting PLV from groups during music composition tasks, although other measures of inter-brain synchrony during musical improvisation tasks, instrumental ensembles, and choir singing are reported [33, 40, 46].

In addition to synchrony, music can also be used as a way to communicate emotion, which is an important part of creating mutual understanding and cooperation between people [53]. Music has even been shown to induce emotional contagion, meaning the emotion expressed by the artist is transferred to the listeners [3, 16]. However, producing music with such an effect requires extensive musical training [32]. Passive BCMI can remove the skill barrier by translating the user’s mental state into music that expresses their emotional valence to listeners with no training or conscious effort. While emotions are complex and hard to classify from neural patterns, Frontal Alpha Asymmetry (FAA) has been evidenced as a robust measure of binary emotional valence [10, 29, 35]. FAA is

defined as the difference between right- and left-hemisphere activity over frontal regions of the brain in the alpha frequency band. Relatively greater alpha activity in the frontal left hemisphere is associated with positive valence and approach motivation, whereas increased alpha activity in the frontal right hemisphere is associated with negative valence and withdrawal motivation [9, 10, 22, 29]. Music has been shown to influence FAA [4] and vice versa, neurofeedback aimed towards modulating FAA to decrease negative valence has been shown to reduce self-reported negative affect, anxiety, and stress [21, 36]. Musical neurofeedback for FAA has not yet been tested but the existing literature is promising especially when combined with evidence of emotional contagion. In a dual-brain system, the musical feedback based on FAA from one user could influence the emotions of another user, resulting in increased empathy and more coordinated emotional states.

3 METHODS

3.1 Music Components and Creation

The BCMI system developed in this project was inspired by the EEG-generated music systems introduced by Wu et al. [54]. In order to produce music, five main parameters needed to be considered: pitch, duration, interval, intensity, and timbre. Pitch relates to the frequency of the note, while duration determines how long the note is played. Interval describes the difference in pitch between two

notes when they are played in sequence or simultaneously. Different intervals between notes result in the combination of these notes sounding either consonant (perceived as pleasant) or dissonant (perceived as jarring). Intensity modulates the volume of the music. Finally, the timbre of a musical note can be described as which instrument the music is produced by: for example, the sample note played by a guitar and piano have different perceptual qualities.

In the proposed BCMI system, three elements of pitch, intensity, and interval of the generated music were determined based on a combination of spectral, temporal, and scale-free features of the live EEG data (Figure 2). More specifically, the mean amplitude, FAA (measuring emotional valence), and PLV (measuring synchrony between participants) were extracted from the EEG signals and mapped to the pitch, intensity, and consonance parameters, respectively. The timbre and duration of the notes were manually modulated through the music production system for the preferred sound effect.

The platform used for generating the music was Sonic Pi [1], a code-based music creation tool that allows for live production of music using MIDI (Musical Instrument Digital Interface) values of notes and a number of parameters to control the musical features described above. MIDI values are essentially used as a digitally transferable representation of pitch. In addition to Sonic Pi, two custom intermediate Python applications are used to process EEG data and convert it to the corresponding musical features, which are then sent to Sonic Pi using a local application of the Open Sound Control (OSC) protocol.

Using Sonic Pi, two tracks of music were played simultaneously; one for each participant. Each track was generated using the corresponding participant's individual EEG features (FAA and scale-free features) as well as the combined PLV value between the participants, resulting in a musical ensemble that represented both the individual mental states of the participants and the synchrony between them.

3.2 EEG Recording and Analysis

3.2.1 EEG hardware. EEG data was collected through two modified Unicorn Hybrid Black headsets (g.tec Medical Engineering, Austria), where the PO7 and PO8 electrodes were replaced with F3 and F4 electrodes. The modified caps had 8 electrodes following the 10/20 universal placement (Fz, F3, F4, C3, Cz, C4, Pz, and Oz) (see Figure 3). The signals were collected at a sampling rate of 250 Hz.

3.2.2 Signal processing. The data processing pipeline began with a 1-35 Hz bandpass filter to clean the data from high-frequency artifacts. Then the first note of the musical composition was determined using the power law mapping rule between pitch and frequency demonstrated by Wu et al. [54] (Equation 1), where Amp is the mean amplitude of the EEG signal, α is the power law exponent index (ranging between 0.5 and 1.5 for EEG data) and n is a constant.

$$\text{Pitch} = (-40/\alpha) \lg \text{Amp} + n \quad (1)$$

After determining the initial note, inter-brain synchrony was used to choose the next notes; higher synchrony was translated into more consonant music and lower synchrony was mapped to

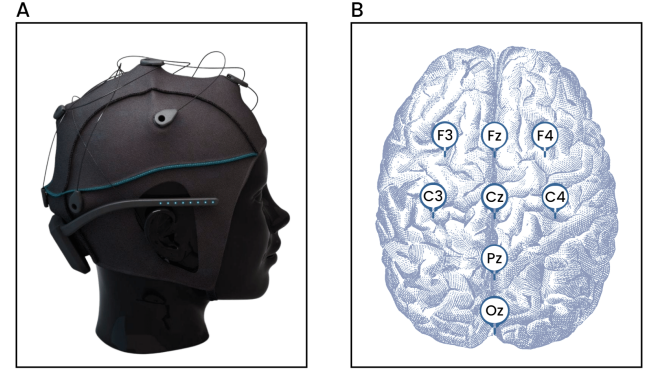


Figure 3: (A) Two Unicorn Hybrid Black headsets (g.tec Medical Engineering) were used for dual EEG recording, (B) EEG signals were collected from 8 electrodes distributed over the scalp.

more dissonant sounds. Starting from the second note, the scale-free note given by the power law conversion was modified to match the closest note that would, depending on the level of synchrony, result in an interval between the two notes that makes them sound consonant in sequence, or one that does the opposite.

Inter-brain Synchrony was calculated using EEG signals from both users. A Hilbert transform (Equation 2) [30] was used to find the PLV between the participants. In this equation, $\theta(t, n)$ is the phase difference between the two signals in the time interval of t for n trials. The obtained PLV value ranges between 0 and 1; 1 indicating absolute synchrony between the two signals and 0 indicating no synchrony. Values indicating a high level of synchronization generated pleasant-sounding combinations of notes, whereas low synchrony would result in dissonant notes that would be audibly jarring when played together.

$$\text{PLV} = \frac{1}{N} \sum_{n=1}^N \exp(j\theta(t, n)) \quad (2)$$

In addition to the pitch and interval features, our system extracted FAA (Equation 3) [9] to determine positive or negative valence from the brain activity of each participant in order to regulate the intensity of the notes. The intensity was changed by adjusting the amplitude of the notes in Sonic Pi, resulting in a higher or lower perceived volume. A negative valence was matched with lower intensity and a positive valence with higher intensity.

$$\text{FAA} = \ln(\alpha_{F3}) - \ln(\alpha_{F4}) \quad (3)$$

3.3 Evaluation

The system has been tested in a laboratory environment on a limited subject pool ($n = 2$) consisting of two of the experimenters (Age = 21 & 22). An example of the resulting audio feed is available here [5]. The distinct audio streams of the two users can be identified as a low-pitched baseline and a higher-pitched melodic layer. In future experiments, it may be interesting to connect more users for a more complex musical piece. The evaluation of this system in

the future may include qualitative measures such as the perceived mood or aesthetic of the music [25], or how ‘in-sync’ users felt with one another. Additionally, it may be interesting to perform a post-analysis of EEG activity to see whether listening to the dual-brain music led to increased synchronicity -or any other connectivity effects- over time.

4 DISCUSSION

The primary goal of this study was to develop a proof of concept for a dual BCMI system that generates ensemble music note-by-note based on both users’ brain activity features. The system extracted three features from both users’ brain activity and sent them to Sonic Pi to generate two tracks of music that were played simultaneously. Each track of music had a different combination of notes and intensity based on the user’s EEG amplitude (converted into scale-free notes using the power-law sonification rule [54]) and emotional valence (measured by Frontal Alpha Asymmetry [9]). Furthermore, both tracks were modulated by the inter-brain synchrony between the participants (measured by Phase-Lock Value [30]). The PLV value was used to alter the intervals between the combined notes, resulting in a more pleasant (consonant) melody for high synchrony and a less pleasant (dissonant) melody for low synchrony. The proposed system could be used as a novel method to provide musical neurofeedback.

The system enables self-monitoring, where team members can use audio feedback to consciously increase their synchrony and individual positive valence, similar to known neurofeedback techniques. Neurofeedback interventions using BCI systems have been shown to improve attention, working memory, and affective states [36, 51]. However, dual-brain BCI systems for neurofeedback of emotion and interpersonal synchrony have not yet been tested in group settings. The expression of emotional and synchrony-based feedback through music could be a useful tool for team members to get on the same wavelength as literature suggests that emotions are contagious [3, 53] and music can both communicate and elicit emotion [16]. Novel methods for group-based neurofeedback could expand the use of neurofeedback beyond the clinical sphere, potentially improving team dynamics and interpersonal cooperation.

Additionally, a passive BCMI system could provide promising applications for general therapeutic insight into individuals’ emotions as well as artistic expression. For example, the ability to convey emotions through music generated by a BCI has potential applications for those with communicative difficulties, such as individuals with a non-verbal Autism Spectrum Disorder, who could benefit from an alternative way to express their emotions and improve coordination with peers. Similarly, for patients with limited muscle control, who would not be able to use traditional musical instruments for creating music, passive BCMI could provide a novel method for artistic and emotional expression. This further suggests the potential of BCMI to become a tool to lower the barrier to musical performance, self-expression, and cooperation between people.

The system described in this paper utilized a commercial EEG cap with a restricted number of 8 electrodes, resulting in a lower spatial resolution compared to EEG acquisition devices with a higher number of electrodes. The usage of an alternative headset may provide additional information from specific brain regions that are critical

for the designated task. Nevertheless, the low cost and wireless nature of the employed EEG cap confers advantages such as ease of use and enhanced user accessibility. A lightweight and portable headset can be crucial, especially in scenarios involving individuals with Autism Spectrum Disorder. A headset that is not bulky and does not require extensive preparation time can improve overall user experience and increase system usability in such cases.

Further research is needed to validate the proposed system in terms of how it is perceived by users, whether they can reliably differentiate between the brain activity-based musical feedback, and whether the feedback has the intended effect. The initial assessment conducted by the researchers suggested that the music quality was satisfactory. However, the ability to draw conclusions about mental states based on the music could not be ascertained without additional evaluation. To determine potential applications of this system, additional experimental studies are needed. However, the literature suggests several promising directions. For instance, musical training within a group setting has been found to have long-term effects on prosocial behaviors such as increased displays of empathy, trust, and cooperation [44, 47]. Additionally, the findings outlined by [8] suggest that adding a measure of interpersonal synchrony, similar to the one developed in this study, would be beneficial for future BCMI systems that aim to improve group dynamics.

Another potential direction for future research could involve exploring the impact of various participant groups on the system outcome. For instance, it may be worthwhile to compare dyads who already have a close relationship to those who are complete strangers. Research has indicated that individuals who are familiar with one another tend to exhibit a higher level of synchrony [27]. Nevertheless, it would be intriguing to examine whether using music as feedback in dual BCIs can elicit a higher level of synchrony between individuals who are unfamiliar with each other.

5 CONCLUSION

A dual brain-computer musical interface (BCMI) was developed to generate live ensemble music from the EEG signals of two users. This system generates music according to the synchrony between two people and each individual’s emotional valence, providing audible feedback of their individual and combined mental states. To create an ensemble, two tracks of music are generated simultaneously, each corresponding to one participant. Music notes are selected per track using a scale-free sonification rule that maps the EEG amplitude to the pitch of the note. The intensity of notes (i.e., the volume of the music) is determined by individuals’ emotional valence and the synchrony between participants determines how consecutive notes are combined; high synchrony produces pleasant-sounding consonant melodies, whereas low synchrony results in dissonant melodies that sound jarring. The musical feedback provided by the system could be used as a novel method for group neurofeedback to increase inter-personal synchrony and subsequently enhance cooperation within teams.

REFERENCES

- [1] Samuel Aaron and Alan F Blackwell. 2013. From sonic Pi to overtone: creative musical experiences with domain-specific and functional languages. In *Proceedings of the first ACM SIGPLAN workshop on Functional art, music, modeling & design*. 35–46.

- [2] Michaël AS Acquadro, Marco Congedo, and Dirk De Ridder. 2016. Music performance as an experimental approach to hyperscanning studies. *Frontiers in human neuroscience* 10 (2016), 242.
- [3] Thomas G Arizmendi. 2011. Linking mechanisms: Emotional contagion, empathy, and imagery. *Psychoanalytic Psychology* 28, 3 (2011), 405.
- [4] Hussain-Abdullah Arjmand, Jesper Hohagen, Bryan Paton, and Nikki S Rickard. 2017. Emotional Responses to Music: Shifts in Frontal Brain Asymmetry Mark Periods of Musical Change. *Frontiers in Psychology* 8 (2017).
- [5] Caterina Ceccato, Ethel Pruss, Anita Vrins, and Jos Prinsen. 2023. *BrainiBeats demo video*. https://www.youtube.com/watch?v=Oh-JXN6_UIE
- [6] Thibault Chabin, Damien Gabriel, Alexandre Comte, Emmanuel Haffen, Thierry Moulin, and Lionel Pazart. 2022. Interbrain emotional connection during music performances is driven by physical proximity and individual traits. *Annals of the New York Academy of Sciences* 1508, 1 (2022), 178–195.
- [7] Rubén Hinojosa Chapel. 2003. Realtime Algorithmic Music Systems From Fractals and Chaotic Functions: Toward an Active Musical Instrument.
- [8] Laura K. Cirelli. 2018. How interpersonal synchrony facilitates early prosocial behavior. *Current opinion in psychology* 20 (2018), 35–39.
- [9] James A Coan and John JB Allen. 2004. Frontal EEG asymmetry as a moderator and mediator of emotion. *Biological psychology* 67, 1-2 (2004), 7–50.
- [10] William A Cunningham, Stacey D Espinet, Colin G DeYoung, and Philip David Zelazo. 2005. Attitudes to the right-and left: frontal ERP asymmetries associated with stimulus valence and processing goals. *NeuroImage* 28, 4 (2005), 827–834.
- [11] Artur Czeszumski, Sara Eustergerling, Anne Lang, David Menrath, Michael Gerstenberger, Susanne Schuberth, Felix Schreiber, Zadkiel Zuluaga Rendon, and Peter König. 2020. Hyperscanning: A Valid Method to Study Neural Inter-brain Underpinnings of Social Interaction. *Frontiers in Human Neuroscience* 14 (2020).
- [12] Ian Daly, Duncan A. H. Williams, Alexis Kirke, James Weaver, Asad Malik, Faustina Hwang, Eduardo Reck Miranda, and Slawomir Jaroslaw Nasuto. 2016. Affective brain-computer music interfacing. *Journal of Neural Engineering* 13 (2016).
- [13] Alexander P. Demos, Roger Chaffin, Kristen T. Begosh, Jennifer R. Daniels, and Kerry L. Marsh. 2012. Rocking to the beat: effects of music and partner's movements on spontaneous interpersonal coordination. *Journal of experimental psychology: General* 141 1 (2012), 49–53.
- [14] Guillaume Dumas, Jacqueline Nadel, Robert Soussignan, Jacques Martinerie, and Line Garnerio. 2010. Inter-brain synchronization during social interaction. *PLoS one* 5, 8 (2010), e12166.
- [15] Joel Eaton, Duncan A. H. Williams, and Eduardo Reck Miranda. 2015. The Space Between Us: Evaluating a multi-user affective brain-computer music interface.
- [16] Hauke Egermann and Stephen McAdams. 2012. Empathy and emotional contagion as a link between recognized and felt emotions in music listening. *Music Perception: An Interdisciplinary Journal* 31, 2 (2012), 139–156.
- [17] Stefan K. Ehrlich, Kat Rose Agres, Cuntai Guan, and Gordon Cheng. 2019. A closed-loop, music-based brain-computer interface for emotion mediation. *PLoS ONE* 14 (2019).
- [18] Walter J Freeman, Mark D Holmes, G Alexander West, and Sampsa Vanhatalo. 2006. Fine spatiotemporal structure of phase in human intracranial EEG. *Clinical Neurophysiology* 117, 6 (2006), 1228–1243.
- [19] Martin Gardner. 1978. Mathematical games: White and brown music, fractal curves and one-over-f fluctuations. *Scientific american* 238, 4 (1978), 16–32.
- [20] Ihshan Gumilar, Ekansh Sareen, Reed Bell, Augustus Stone, Ashkan Hayati, Jingwen Mao, Amit Barde, Anubha Gupta, Arindam Dey, Gun Lee, et al. 2021. A comparative study on inter-brain synchrony in real and virtual environments using hyperscanning. *Computers & Graphics* 94 (2021), 62–75.
- [21] Yasir Hafeez, Syed Saad Azhar Ali, Rumaisa Abu Hasan, Syed Hasan Adil, Muhammad Moinuddin, Mansoor Ebrahim, Muhammad Saiful Bahri Yusoff, Hafeezullah Amin, and Ubaid Al-Saggaf. 2021. Development of Enhanced Stimulus Content to Improve the Treatment Efficacy of EEG-Based Frontal Alpha Asymmetry Neurofeedback for Stress Mitigation. *IEEE Access* 9 (2021), 130638–130648.
- [22] Eddie Harmon-Jones. 2003. Early Career Award. Clarifying the emotive functions of asymmetrical frontal cortical activity. *Psychophysiology* 40 6 (2003), 838–848.
- [23] Thilo Hinterberger and Gerold Baier. 2005. Parametric orchestral sonification of EEG in real time. *IEEE MultiMedia* 12, 2 (2005), 70–79.
- [24] Yinying Hu, Min Zhu, Yang Liu, Zixuan Wang, Xiaojun Cheng, Yafeng Pan, and Y. Hu. 2022. Musical Meter Induces Interbrain Synchronization during Interpersonal Coordination. *eNeuro* 9 (2022).
- [25] Eva Istók, Elvira Brattico, Thomas Jacobsen, Kaisu Krohn, Mira Müller, and Mari Tervaniemi. 2009. Aesthetic responses to music: A questionnaire study. *Musicae Scientiae* 13, 2 (2009), 183–206.
- [26] Peter E. Keller, Giacomo Novembre, and Michael J. Hove. 2014. Rhythm in joint action: psychological and neurophysiological mechanisms for real-time interpersonal coordination. *Philosophical Transactions of the Royal Society B: Biological Sciences* 369 (2014).
- [27] Sivan Kinreich, Amir Djalovski, Lior Kraus, Yoram Louzoun, and Ruth Feldman. 2017. Brain-to-brain synchrony during naturalistic social interactions. *Scientific reports* 7, 1 (2017), 17060.
- [28] Ivana Konvalinka, Markus Bauer, Carsten Stahlhut, Lars Kai Hansen, Andreas Roepstorff, and Chris D. Frith. 2014. Frontal alpha oscillations distinguish leaders from followers: Multivariate decoding of mutually interacting brains. *NeuroImage* 94 (2014), 79–88.
- [29] Claudia Krogmeier, Brandon S Coventry, and Christos Mousas. 2022. Affective Image Sequence Viewing in Virtual Reality Theater Environment: Frontal Alpha Asymmetry Responses From Mobile EEG. *Frontiers in Virtual Reality* (2022).
- [30] Jean-Philippe Lachaux, Eugenio Rodriguez, Jacques Martinerie, and Francisco J. Varela. 1999. Measuring phase synchrony in brain signals. *Human Brain Mapping* 8 (1999).
- [31] Sylvain Le Groux, Jonatas Manzolli, Paul FMJ Verschure, Marti Sanchez, Andre Luvizotto, Anna Mura, Aleksander Valjamae, Christoph Guger, Robert Prueckl, and Ulysses Bernardet. 2010. Disembodied and Collaborative Musical Interaction in the Multimodal Brain Orchestra. In *NIME*. Citeseer, 309–314.
- [32] Chih Yi Lin and Stone Cheng. 2012. Multi-theme analysis of music emotion similarity for jukebox application. *2012 International Conference on Audio, Language and Image Processing* (2012), 241–246.
- [33] Ulman Lindenberger, Shu-Chen Li, Walter Gruber, and Viktor Müller. 2009. Brains swinging in concert: cortical phase synchronization while playing guitar. *BMC neuroscience* 10, 1 (2009), 1–12.
- [34] Bill Manaris, Juan Romero, Penousal Machado, Dwight Krehbiel, Timothy Hirzel, Walter Pharr, and Robert B Davis. 2005. Zipf's law, music classification, and aesthetics. *Computer Music Journal* 29, 1 (2005), 55–69.
- [35] Danielle Mathersul, Leanne M Williams, Patrick J Hopkinson, and Andrew H Kemp. 2008. Investigating models of affect: relationships among EEG alpha asymmetry, depression, and anxiety. *Emotion* 8, 4 (2008), 560.
- [36] Rocco Mennella, Elisabetta Patron, and Daniela Palomba. 2017. Frontal alpha asymmetry neurofeedback for the reduction of negative affect and anxiety. *Behaviour research and therapy* 92 (2017), 32–40.
- [37] Eduardo Reck Miranda and Julien Castet. 2014. Guide to Brain-Computer Music Interfacing. In *Springer London*.
- [38] Evan Morgan, Hatice Gunes, and Nick Bryan-Kinns. 2015. Using affective and behavioural sensors to explore aspects of collaborative music making. *International Journal of Human-Computer Studies* 82 (2015), 31–47.
- [39] Tim R. Mullen, Alexander Khalil, Tomas E. Ward, John Rehner Iversen, Grace Leslie, Richard Warp, Matt Whitman, Victor Minces, Aaron McCoy, Alejandro Ojeda, Nima Bigdely-Shamlo, Mike Chi, and David Rosenboom. 2015. MindMusic: Playful and Social Installations at the Interface Between Music and the Brain.
- [40] Viktor Müller, Johanna C. Sanger, and Ulman Lindenberger. 2018. Hyperbrain network properties of guitarists playing in quartet. *Annals of the New York Academy of Sciences* 1423 (2018).
- [41] Anton Nijholt. 2015. Competing and collaborating brains: multi-brain computer interfacing. In *Brain-Computer Interfaces*. Springer, 313–335.
- [42] Naoyuki Osaka, Takehiro Minamoto, Ken Yai, Miyuki Azuma, Yohko Minamoto Shimada, and Mariko Osaka. 2015. How two brains make one synchronized mind in the inferior frontal cortex: fNIRS-based hyperscanning during cooperative singing. *Frontiers in psychology* 6 (2015), 1811.
- [43] Andreas Pinegger, Hannah Hiebel, Selina C Wriessnegger, and Gernot R Müller-Putz. 2017. Composing only by thought: Novel application of the P300 brain-computer interface. *PLoS one* 12, 9 (2017), e0181584.
- [44] Tal-Chen Rabinowitch, Ian Cross, and Pamela Burnard. 2013. Long-term musical group interaction has a positive influence on empathy in children. *Psychology of Music* 41 (2013), 484 – 498.
- [45] Mauricio A Ramirez-Moreno, Jesús G Cruz-Garza, Akanksha Acharya, Girija Chatufale, Woody Witt, Dan Gelok, Guillermo Reza, and José L Contreras-Vidal. 2022. Brain-to-brain communication during musical improvisation: a performance case study. *F1000Research* 11, 989 (2022), 989.
- [46] Johanna Sanger, Viktor Müller, and Ulman Lindenberger. 2012. Intra- and interbrain synchronization and network properties when playing guitar in duets. *Frontiers in human neuroscience* (2012), 312.
- [47] E. Glenn Schellenberg, Kathleen A. Corrigan, Sebastian P Dys, and Tina Malti. 2015. Group Music Training and Children's Prosocial Skills. *PLoS ONE* 10 (2015).
- [48] Benjamin T. Schmidt, Avniel Singh Ghuman, and Theodore J. Huppert. 2014. Whole brain functional connectivity using phase locking measures of resting state magnetoencephalography. *Frontiers in Neuroscience* 8 (2014).
- [49] Davide Valeriani and Ana Matran-Fernandez. 2018. Past and future of multi-mind brain-computer interfaces. *Brain-Computer Interfaces Handbook* (2018), 685–700.
- [50] Satvik Venkatesh, Eduardo Reck Miranda, and Edward Braund. 2022. SSVEP-based brain-computer interface for music using a low-density EEG system. *Assistive Technology* (2022), 1–11.
- [51] Jinn-Rong Wang and Shulan Hsieh. 2013. Neurofeedback training improves attention and working memory performance. *Clinical Neurophysiology* 124 (2013), 2406–2420.
- [52] Valtteri Wikström, Katri Saarikivi, Mari Falcon, Tommi Makkonen, Silja Martikainen, Vesa Putkinen, Benjamin Ultan Cowley, and Mari Tervaniemi. 2022. Inter-brain synchronization occurs without physical co-presence during cooperative online gaming. *Neuropsychologia* 174 (2022).

- [53] Duncan A. H. Williams. 2019. Evaluating BCI for Musical Expression: Historical Approaches, Challenges and Benefits. In *Brain Art*.
- [54] Dan Wu, Chaoyi Li, Jie Liu, Jing Lu, and Dezhong Yao. 2014. Scale-free brain ensemble modulated by phase synchronization. *Journal of Zhejiang University SCIENCE C* 15 (2014), 821–831.
- [55] Kyongsik Yun, Katsumi Watanabe, and Shinsuke Shimojo. 2012. Interpersonal body and neural synchronization as a marker of implicit social interaction. *Scientific reports* 2, 1 (2012), 1–8.
- [56] Wei Zhou, Chenyang Qiu, and Guangyuan Liu. 2021. Efficient regulation of emotion by positive music based on EEG valence-arousal model. *2021 3rd International Conference on Image, Video and Signal Processing* (2021).

Received 19 January 2023